

3.1.4 Energetics

The enthalpy change in a chemical reaction can be measured accurately. It is important to know this value for chemical reactions that are used as a source of heat energy in applications such as domestic boilers and internal combustion engines.

3.1.4.1 Enthalpy change

Content	Opportunities for skills development
Reactions can be endothermic or exothermic. Enthalpy change (ΔH) is the heat energy change measured under conditions of constant pressure. Standard enthalpy changes refer to standard conditions, ie 100 kPa and a stated temperature (eg $\Delta H_{298}^{\ominus}$). Students should be able to: • define standard enthalpy of combustion ($\Delta_c H^{\ominus}$) • define standard enthalpy of formation ($\Delta_f H^{\ominus}$).	

3.1.4.2 Calorimetry

Content	Opportunities for skills development
The heat change, q, in a reaction is given by the equation $q = mc\Delta T$ where m is the mass of the substance that has a temperature change ΔT and a specific heat capacity c. Students should be able to: • use this equation to calculate the molar enthalpy change for a reaction • use this equation in related calculations. Students will not be expected to recall the value of the specific heat capacity, c, of a substance.	MS 0.0 and 1.1 Students understand that the correct units need to be used in $q = mc\Delta T$ Students report calculations to an appropriate number of significant figures, given raw data quoted to varying numbers of significant figures. Students understand that calculated results can only be reported to the limits of the least accurate measurement.
Required practical 2 Measurement of an enthalpy change.	AT a and k PS 2.4, 3.1, 3.2, 3.3 and 4.1 Students could be asked to find ΔH for a reaction by calorimetry. Examples of reactions could include: • dissolution of potassium chloride • dissolution of sodium carbonate • neutralising NaOH with HCl • displacement reaction between $\text{CuSO}_4 + \text{Zn}$ • combustion of alcohols.

3.1.4.3 Applications of Hess's law

Content	Opportunities for skills development
Hess's law. Students should be able to: use Hess's law to perform calculations, including calculation of enthalpy changes for reactions from enthalpies of combustion or from enthalpies of formation.	MS 2.4 Students carry out Hess's law calculations. AT a and k PS 2.4, 3.2 and 4.1 Students could be asked to find ΔH for a reaction using Hess's law and calorimetry, then present data in appropriate ways. Examples of reactions could include: - thermal decomposition of NaHCO_3 - hydration of MgSO_4 - hydration of CuSO_4

3.1.4.4 Bond enthalpies

Content	Opportunities for skills development
Mean bond enthalpy. Students should be able to: - define the term mean bond enthalpy - use mean bond enthalpies to calculate an approximate value of ΔH for reactions in the gaseous phase - explain why values from mean bond enthalpy calculations differ from those determined using Hess's law.	MS 1.2 Students understand that bond enthalpies are mean values across a range of compounds containing that bond.